

HOMEWORK 3

Problem 1

Given the provided dataset “HW3_data.csv”, consider the model regressing y on x and m . Conduct a Sobel test to see whether there is a mediation effect. You need to consider the three relevant models,

$$y_i = \beta_{11} + \beta_{12}x_i + \epsilon_i,$$

$$m_i = \beta_{21} + \beta_{22}x_i + \epsilon_i,$$

and

$$y_i = \beta_{31} + \beta_{32}m_i + \beta_{33}x_i + \epsilon_i.$$

Suppose the variance of ϵ_i is known with $\sigma^2 = 1$ for all the three models. Use the level of significance $\alpha = 0.05$

- (1) (4 pts) What is the null hypothesis of the Sobel test?
- (2) (8 pts) Regress the three models and find the estimators for (β_{11}, β_{12}) , (β_{21}, β_{22}) and $(\beta_{31}, \beta_{32}, \beta_{33})$.
- (3) (10 pts) Find the variance of $\hat{\beta}_{22}$ and $\hat{\beta}_{32}$.
- (4) (10 pts) Find the z test statistic.
- (5) (8 pts) Find the p -value and the conclusion of the test.

Problem 2

Consider the usual simple linear model (i.e. there is only a single covariate and hence only a single unknown β , assuming the data is centered, so no intercept term),

$$y_i = x_i\beta + \sigma\epsilon_i,$$

where σ is unknown and $\sum_{i=1}^n x_i^2 = n$.

- (1) (12 pts) Show that the least square estimator is given by $\hat{\beta} = n^{-1} \sum_{i=1}^n x_i y_i$.
- (2) (12 pts) Show that the residual sum of squares is given by $\sum_{i=1}^n y_i^2 - n\hat{\beta}^2$.
- (3) (12 pts) Find the F statistic for testing $H_0 : \beta = 0$
- (4) (12 pts) If the null hypothesis is true, what is the distribution of $\hat{\beta}$?
- (5) (12 pts) Hence, show that the Student- t test statistic for testing the null hypothesis is

$$T = \sqrt{n}\hat{\beta} / \hat{\sigma}, \quad \text{with} \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

and confirm that the F test and the T test are the same.